FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

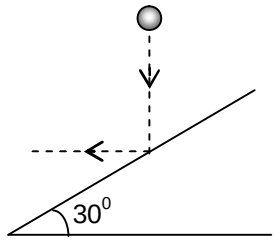
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

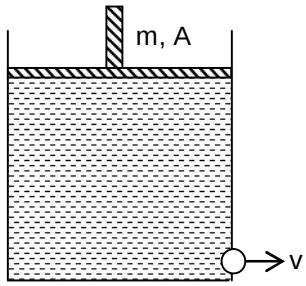
1. A ball is released form a point, it goes vertically downwards and collides with a fixed smooth inclined plane of angle of inclination of 30^0. After the collision the ball goes horizontally. The coefficient of restitution between the ball and the inclined plane is



(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) 1 (D) None of these

2. A cylindrical vessel contains a liquid of density ρ up to height h. The liquid is closed by a piston of mass m and area of cross section A. There is a small hole at the bottom of the vessel. The speed v with which the liquid comes out of the hole is



(A) $\sqrt{2gh}$

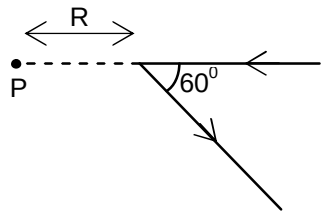
(B) $\sqrt{2\left(gh+\frac{mg}{\rho A}\right)}$

(C) $\sqrt{2\left(gh+\frac{mg}{A}\right)}$

(D) $\sqrt{2gh+\frac{mg}{A}}$

Space for Rough work

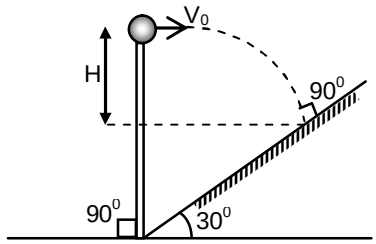
3. A long straight wire, carrying a current I is bent at its mid point to form an angle of 60°. AT a point P, distance R from the point of bending the magnetic field is



(A) $\frac{\left(\sqrt{2}-1\right)\mu_0 i}{4\pi R}$ (B) $\frac{\left(\sqrt{2}+1\right)\mu_0 i}{4\pi R}$

(C) $\frac{\mu_0 i}{4\sqrt{3}\pi R}$ (D) $\frac{\mu_0 i}{8R}$

4. In the given figure, the angle of inclination of the inclined plane is 30^0. The horizontal velocity V_0 so that the particle hits the inclined plane perpendicularly is



(A) $v_0 = \sqrt{\frac{2gH}{3}}$

(B) $V_0 = \sqrt{\frac{2gH}{7}}$

(C) $V_0 = \sqrt{\frac{gH}{5}}$

(D) $V_0 = \sqrt{\frac{gH}{7}}$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. In the series L – C – R circuit, the voltage across resistance, capacitance and inductance are 30V each at frequency $f = f_0$.

(A) It the inductor is short-circuited, the voltage across the capacitor will be $30\sqrt{2}$ V.

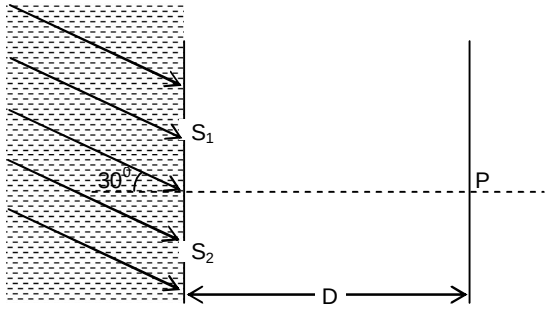
(B) If the capacitor is short-circuited, the voltage drop across the inductor will be $\frac{30}{\sqrt{2}}$ V.

(C) If the frequency is changed to $2f_0$, the ratio of reactance of the inductor to that of the capacitor is 4 : 1.

(D) If the frequency is changed to $2f_0$, the ratio of the reactance of the inductor to that of the capacitor is 1 : 4.

Space for Rough work

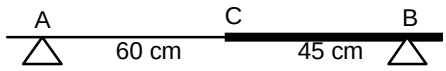
6. The given figure shows a YDSE apparatus. Parallel monochromatic coherent light rays are incident on slits S_1 and S_2 ($S_1S_2 = \frac{2}{3}$mm) at an angle 30° with the horizontal. The medium on left side of the slits is water $(\mu_w = 4/3)$. To obtain the central maxima at point P, a glass slab $(\mu_g = 3/2)$ inside water is introduced in front of slit S_1. The thickness of the glass slab required for this purpose is



(A) 2mm (B) 4/3 mm
(C) 4 mm (D) 8/3 mm

7. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A eV and de-Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B=(T_A - 1.50)$ eV. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$ then
(A) the work function of A is 2.25 eV (B) the work function of B is 4.20 eV
(C) T_A = 2.00 eV (D) T_B = 2.75 eV

8. A steel wire of length 60 cm and area of cross section $10^{-6}m^2$ is joined with an aluminium wire of length 45 cm and area of cross section $3\times10^{-6}m^2$. The composite string is stretched by a tension of 80 N. Density of steel is 7800 kg m^{-3} and that of aluminium is 2600 kg m^{-3}. The minimum frequency of turning fork, which can produce standing wave in it with node at joint is



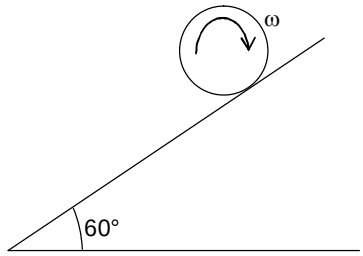
(A) 357.3 Hz (B) 375.3 Hz
(C) 337.5 Hz (D) 325.3 Hz

Space for Rough work

9. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg. After their collision, the 1 kg mass reverses its direction and moves with a speed of 2 ms^{-1}. Which of the following statement(s) is (are) correct for the system of these two masses?
(A) Total momentum of the system is 3 kg ms^{-1}
(B) Momentum of 5 kg mass after collision is 4 kg ms^{-1}
(C) Kinetic energy of the centre of mass is 0.75 J
(D) Total kinetic energy of the system is 4 J

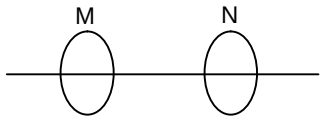
10. A strip of wood of mass M and length ℓ is placed on a smooth horizontal surface. An insect of mass m starts at one end on the strip and walks to the other end in time t, moving with a constant speed.
(A) the speed of the insect as seen from the ground is $< \frac{\ell}{t}$.
(B) the speed of the strip as seen from the ground is $\frac{\ell}{t}\left(\frac{M}{M+m}\right)$
(C) the speed of the strip as seen from the ground is $\frac{\ell}{t}\left(\frac{m}{M+m}\right)$
(D) the total kinetic energy of the system is $\frac{1}{2}(m+M)\left(\frac{\ell}{t}\right)^2$

11. A cylinder is rotated clockwise and lowered slowly on a rough inclined plane with (μ = 0.8). Then:
(A) cylinder will start going upwards
(B) cylinder will start going downwards.
(C) frictional force will act upwards.
(D) frictional force will act downwards.



Space for Rough work

12. Two identical circular coils M and N are arranged coaxially as shown in the figure. Separation between the coils is large as compared to their radii. The arrangement is viewed from left along the common axis. The sign convention adopted is that currents are taken to be positive when they appear to flow in clockwise direction. Then



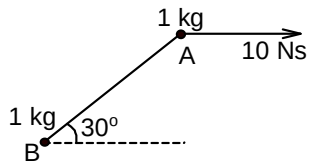
(A) if M carries a constant positive current and is moved towards N, a positive current is induced in N
(B) if M carries a constant positive current and N is moved towards M, a negative current is induced in N
(C) if a positive current in M is switched off, a positive current is momentarily induced in N
(D) if both coils carry positive currents, they will attract each other

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

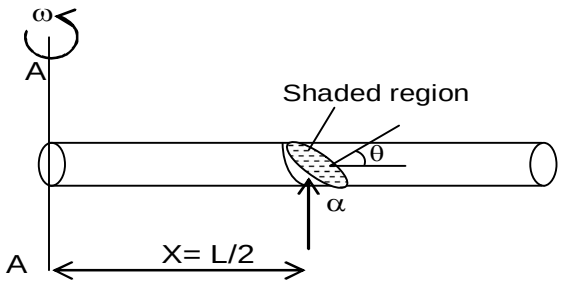
13. A rocket of mass 40kg has 360 kg of fuel. The exhaust velocity of the fuel is 2.0 km/sec. Calculate minimum rate of consumption of fuel in kg/s so that the rocket may rise from the ground (g = 10 m/sec2).

14. Two balls A and B of masses 1 kg each are connected by an inextensible massless string. The system is resting on a smooth horizontal surface. An impulse of 10 Ns is applied to the ball A at an angle 30° with the line joining two balls in horizontal direction as shown in the figure. Assuming that the string remains taut after the impulse, if the magnitude of kinetic energy of ball A is $\frac{175}{k}$. Find the value of K.



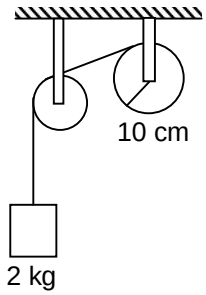
Space for Rough work

15. A thin uniform metallic rod of mass m, length L, Young modulus of elasticity Y and cross–sectional area A is rotated by angular velocity ω about extreme end AA'. Consider a section on the rod at midpoint of rod. What will be the normal stress in N/m^2 on the shaded region? (Take the value of $\frac{M\omega^2L^2\cos^2\theta}{A} = 10\,N/m^2$)

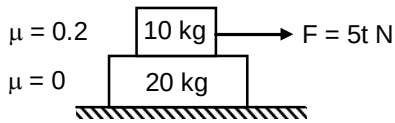


16. The thickness of a glass plate is measured to be 2.17 mm, 2.17 mm and 2.18 mm at three different places. Find the average thickness of the plate from this data. (in mm).

17. A string is wrapped on a wheel of moment of inertia 0.20 kg m^2 and radius 10 cm and goes through a light pulley to support a block of mass 2.0 kg as shown in figure. Find the acceleration of the block. (in m/s^2)



18. Find the speed of 10 kg block at the end of 3 second.



Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. In a photoelectric experiment with a particular incident radiation, the emission of the number of photoelectrons depends on
(A) the frequency of incident radiation (B) the intensity of incident radiation
(C) the wavelength of incident radiation (D) the wave number of incident radiation

20. The most thermally unstable complex is
(A) $[Cr(NH_3)_6]\ [Co(CN)_6]$ (B) $[Cr(NH_3)_2(CN)_4][Co(NH_3)_4(CN)_2]$
(C) $[Cr(NH_3)_3(CN)_3]\ [Co(NH_3)_3(CN)_3]$ (D) $[Cr(NH_3)(CN)_5][Co(NH_3)_5(CN)]$

21. If Λ_m = molar conductivity of the solution of an electrolyte.
Λ_m^0 = limiting molar conductivity of the solution
α = degree of dissociation of the electrolyte,
Then the relation among the three terms is
(A) $\frac{\Lambda_m^0}{\Lambda_m} = \alpha$ (B) $\frac{\Lambda_m}{\Lambda_m^0} = \alpha$
(C) $\Lambda_m^0 \times \Lambda_m = \alpha$ (D) $\Lambda_m^0 - \Lambda_m = \alpha$

22. P_4O_{10} can react with
(A) HCl (B) HNO_3
(C) HBr (D) HI

Space for Rough work

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

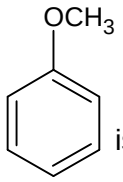
23. The correct statement(s) regarding CH_4 is
(A) it is a tetrahedral molecule
(B) it undergoes substitution reaction
(C) it can be formed by the reaction between CH_3MgBr and phenol
(D) it diffuses at a faster rate than dioxygen

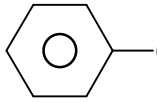
24. Which of the following compound(s) is/are more basic than CH_3NH_2 in gaseous state?
(A) CH_3NHCH_3 (B) $(CH_3)_3N$
(C) CH_3CN (D) CH_3NO_2

25. $A(s) + 2B(g) \rightleftharpoons C(g)$
Correct statement(s) are
(A) the reaction will proceed towards forward direction by increasing pressure.
(B) if the concentrations of B and C are halved at equilibrium, then the reaction will proceed towards backward direction
(C) addition of A makes the reaction proceed towards forward direction
(D) if the volume of reaction containers are halved, the reaction will proceed towards forward direction.

26. Which of the following substances on heating produces PH_3?
(A) PH_4Cl (B) H_3PO_3
(C) H_3PO_4 (D) Ca_3P_2

Space for Rough work

27. The correct statement(s) regarding anisole  is/are

(A) it can be formed by treating phenol with hot CH_2N_2.
(B) it can undergo halogenation reaction(both electrophilic and free radical mechanism).
(C) it is oxidized to benzoic acid.
(D) it can be produced by treating  with CH_3Br.

28. $CN^- + H_2O \rightleftharpoons HCN + OH^-$

Addition of which of the following reagent can make the reaction proceed towards forward direction.

(A) H_2O (B) HCl
(C) NaBr (D) KOH

29. Which wave functions(ψ) do/does not represent any atomic orbital?

(A) $\psi_{2,1,2}$ (B) $\psi_{3,0,0}$
(C) $\psi_{4,2,-1}$ (D) $\psi_{3,3,1}$

30. Which of the following factor(s) increase(s) the rate of a zero order reaction?
(A) increasing temperature
(B) increasing reactant concentration
(C) decreasing half-life period
(D) increasing pressure for gaseous reactants

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

31. $2A(s) \rightleftharpoons B(g) + 2C(g)$

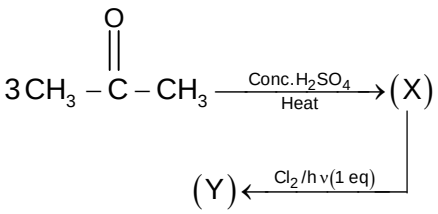
Above reaction attains equilibrium at 240^oC and 19.5 atm pressure when initially only A(s) is taken for reaction. What is the equilibrium constant K_P of the reaction?

32. 'X' is an aliphatic monocarboxylic acid. It is the simplest acid which can form only one alpha(α)-chloroproduct(Y), upon treatment with Cl_2 in presence of red phosphorus (H.V.Z reaction). What is the molar mass of Y in g mol^{-1} unit?

33. AB is a polar molecule It's bond length is $5\,\mathring{A}$. If the maximum dipole moment of the molecule is $b \times 10^{-29}$ coulomb m, what is the value of 'b'? [charge of e^- = 1.602×10^{-19} C]

34. One litre aqueous solution contains 0.1 mole of a weak monobasic acid HA and 1 mole of its sodium salt NaA. If the pH of the solution is 6.49 what is the pK_a of HA?

35.



What is the molar mass of compound (Y) in g mol^{-1} unit?

36. Salt(A) produces golden yellow flame on flame test. Aqueous solution of (A) produces a white precipitate with barium chloride solution. The precipitate is insoluble in dilute mineral acids. On thermal decomposition in presence of SiO_2, (A) forms a salt (B) and gas(C). When treated with an acid(D), salt(B) forms NaCl, SiO_2 and H_2O. What is the molar mass of (D)?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. The number of values of k for which the equation $x^3 - 3x + k = 0$ has two distinct roots lying in the interval (0, 1) is
(A) three (B) two
(C) infinitely many (D) zero

38. If $f : A \to B$, where $A = \{a,b,c,d\}$ and $B = \{x,y,z\}$ and $f(a) = x$ then number of such INTO functions is
(A) 27 (B) 15
(C) 21 (D) none of these

39. If α is a root of $x^2 + ax + 1 = 0$, then $\lim\limits_{x \to 1/\alpha} \dfrac{\sin(x^2 + ax + 1)}{(\alpha x - 1)}$ is equal to
(A) $2a\alpha$ (B) $a\alpha^2$
(C) $\dfrac{1-\alpha^2}{\alpha^2}$ (D) does not exist

40. If $f(x) = x^4 \tan(x^3) - x \ln(1 + x^2)$, then the value of $\dfrac{d^4(f(x))}{dx^4}$ at x = 0 is
(A) 0 (B) 6
(C) 12 (D) 24

Space for Rough work

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. Which of the following is /are true ?

(A) $\int_0^1 \sin\left(x^2+2x+1\right)dx - \int_1^2 \sin x^2 dx = 0$

(B) $\int_{-1}^1 e^{\sin x}dx - \int_0^1 e^{\sin(2x+1)}dx = 0$

(C) $\int_{-5}^{-4} \sin\left(x^2-3\right)dx + \int_{-2}^{-1} \sin\left(x^2+12x+33\right)dx = 0$

(D) $\int_{-4}^4 \cos x^2\, dx - 8\int_0^1 \cos 16(2x-1)^2 dx = 0$

42. If A and B are two events such that $P(A) = 3/4$ and $P(B) = 5/8$, then

(A) $P(A \cup B) \geq 3/4$ (B) $P(A' \cap B) \leq 1/4$

(C) $3/8 \leq P(A \cap B) \leq 5/8$ (D) $3/8 \leq P(A \cap B) \leq 5/8$

43. Consider a sequence $\{a_n\}$ with $a_1 = 2$ and $a_n = \frac{a_{n-1}^2}{a_{n-2}}$ for all $n \geq 3$, terms of the sequence being distinct. Given that a_2 and a_5 are positive integers and $a_5 \leq 162$ then the possible value(s) of a_5 can be

(A) 162 (B) 64

(C) 32 (D) 2

Space for Rough work

44. Consider the planes $3x-6y+2z+5=0$ and $4x-12y+3z=3$. The plane $67x-162y+47z+44=0$ bisects the angle between the given planes which

(A) contains the origin
(B) is acute
(C) is obtuse
(D) none of these

45. If h_1, h_2, h_3 are length of altitudes of a triangle and $\frac{1}{h_1}+\frac{1}{h_2}+\frac{1}{h_3}=\frac{2}{R}$, where R is the radius of its circumcircle, then

(A) $2h_1=3k$
(B) $3h_1=2R$
(C) $2h_2=3R$
(D) $3h_2=2R$

46. A vector of magnitude $\sqrt{29}$ which makes equal angles with vectors $\hat{i}+\hat{j}-2\hat{k}, 2\hat{i}-\hat{j}+3\hat{k}, \hat{i}-\hat{j}+\hat{k}$ is $a\hat{i}+b\hat{j}+c\hat{k}$ then $a+b+c$ can be

(A) 1
(B) 2
(C) –2
(D) –1

47. If $f(x)=\int_0^x \frac{\sin t}{t}dt,\ x>0$ then

(A) $f(x)$ has a local maxima at $x=n\pi\left(n=2k, k\in I^+\right)$
(B) $f(x)$ has a local minima at $x=n\pi\left(n=2k, k\in I^+\right)$
(C) $f(x)$ has neither maxima nor minima at $x=n\pi\left(n\in I^+\right)$
(D) $f(x)$ has local maxima at $x=n\pi\left(n=2k-1, k\in I^+\right)$

48. If $x\in R$, then the roots of the equation $x^4+4x^3-8x^2+k=0$ with respect to the values of k, $\left(k\in[0,2014]\right)$, are :

(A) for $k\in[0,3]$: 4 real and distinct roots
(B) for $k\in(3,128)$: two real and two imaginary roots
(C) for $k\in(128,583)$: no real root
(D) for $k\in(0,3)$: four real and distinct roots

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

49. Let $a_1, a_2, a_3 ....., a_{4001}$ are in A.P such that $\frac{1}{a_1a_2} + \frac{1}{a_2a_3} + ... + \frac{1}{a_{4000}a_{4001}} = 10$ and $a_2 + a_{4000} = 50$ then $|a_1 - a_{4001}|$ is equal to

50. If $|z - 3| = \text{Re } z$, $|w - 3| = \text{Re } w$ and $\arg(z - w) = \frac{\pi}{4}$, then find the value of $\text{Im}(z + w)$.

51. The number of functions from $f : A \to B$ where $A = \{a_1, a_2, a_3, a_4, a_5, a_6\}$, $B = \{1, 2, 3, \ldots, 9\}$ such that $f(a_{i+1}) > f(a_i)\ \forall\ i \in N$ and $f(a_i) \neq i$ is

52. Let $(x^3 + px^2 + 2x - 5)^{19} (x^2 + qx - 41)^8 (x^4 - x^3 + x - 7)^6 = x^{97} + 391x^{96} + a_{95}x^{95} + a_{94}x^{94} + ... a_1x + a_0$ be an identity, where p, q, $a_{95}....a_0$ are integers. Compute the smallest positive value of p.

53. $\vec{a}, \vec{b}$ and $\vec{c}$ be three non – coplanar vectors, and $\vec{d}$ be a non zero vector, which is perpendicular to $(\vec{a} + \vec{b} + \vec{c})$. Now if $\vec{d} = \sin x(\vec{a} \times \vec{b}) + \cos y(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})$, then minimum value of $x^2 + y^2$ is equal to $\frac{\lambda\pi^2}{4}$, where λ is

54. If $\int \frac{(x-1)dx}{(x + x\sqrt{x} + \sqrt{x})\sqrt{\sqrt{x}(x+1)}} = 4\tan^{-1}[g(x)] + C$, where C is an arbitrary constant of integration. Find $g^2(1)$.

Space for Rough work